

Does a Slime Mold Think?

What counts as cognition — and does biology matter?

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drafted with Claude (Anthropic)

Computational Modelling for Psychology & Neuroscience

Where we've been

- ▶ **Dog · bell · food** — nobody argues. We grant “it learned.”
- ▶ **Sunflower · sun · sugar** — anticipation of dawn, but *built in*. Not learned.
- ▶ **Pea plant · fan · light** — *claimed* associative learning. Contested; a replication failed.

A single cell with no brain

Physarum polycephalum — a slime mold.

- ▶ One giant **multinucleate cell**. No neurons, no synapses, no nervous system.
- ▶ Forages by growing a living network of tubes; routes are reinforced or pruned by internal **flow feedback**.
- ▶ Long filed under “fungus”; actually an amoeba.

Everything on the next slides is done by *this*.

It solves mazes

Place it between two food sources in a maze. It fills the maze, then **withdraws from dead ends** and keeps the tube along the **shortest path** connecting the food.

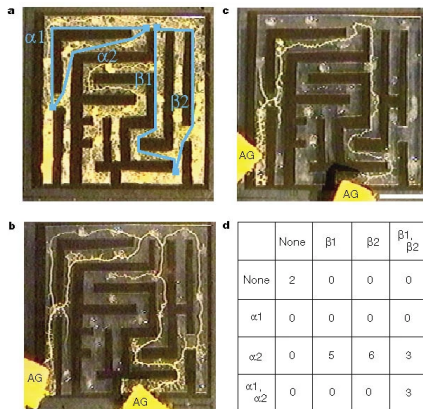


Figure 1: Maze Navigation in the Slime Mold

It designs networks that rival engineers

Arrange oat flakes as the cities of Greater Tokyo. The mold grows a network that **approximates the real rail system** — balancing total tube length, transport efficiency, and **fault tolerance** against accidental breaks.

Those are the exact tradeoffs human planners spend months negotiating.

— *Tero et al. (2010), Science*

But is it *learning*?

The sunflower failed this test. The slime mold does not.

- ▶ **Habituation** — made to cross a bridge laced with a *harmless* bitter substance, it hesitates at first, then stops reacting over days — and recovers if left alone. Habituation is a textbook elementary form of learning.
- ▶ **Anticipation** — shocked with cold at fixed intervals, it slows *before* the next shock — and still slows once at the expected time when the shock is withheld.

— *Boisseau, Vogel & Dussutour (2016); Saigusa et al. (2008)*

Memory — without a brain ...

and outside the body

It avoids ground it has already covered by **sensing the extracellular slime it left behind**. The record of where it has been is written into the *environment*, not stored in a head.

— Reid, Latty, Dussutour & Beaton (2012), PNAS

So: Does it Think?

Which definition are you using?

- ▶ **Functional** — cognition is what a system *does* (solve, learn, remember). → It qualifies.
- ▶ **Substrate** — cognition is a property of certain *stuff* / *architecture* (nervous systems). → It doesn't.

Is a mathematical model of the slime mold computational cognitive (neuro)science?

A mathematical model of slime mold shortest path computation.

Math Model

Can an organism *be* an algorithm?

To answer “does the slime mold think?” do you have to decide if **the computation it runs is cognition?**

— *Tero et al. (2010); Bonifaci, Mehlhorn & Varma (2012)*

Our four test cases compared

“X learns that Y predicts Z” hides four separate claims. Line the cases up against them:

Does the case have. . .	Dog	Sunflower	Pea	Slime mold	LLM
a real $Y \rightarrow Z$	yes	yes	yes	yes	yes
contin- gency behaviour that tracks it	yes	yes	<i>claimed</i>	yes	yes
acquisition by experience	yes	no (built-in)	<i>claimed</i>	yes	train: yes / run: no
an internal representa- tion	yes?	a clock?	<i>claimed</i>	external?	<i>contested</i>

- ▶ The **top two rows are trivial**. A thermostat passes.
- ▶ **Acquisition** is *learning*. Is the sunflower learning? No. The dog and slime mold? Yes.
- ▶ **Representation** — this is where we assume people live and where cognition is defined. But then don't we have to grant the pea plant and slime mold cognitive status?

The Assignment

Write your answer to the question,

“Does a slime mold think?”

One page maximum. As long as I can read it you are good. No worries about spelling or grammar, but don't be careless. I am not looking for a particular answer. I will be reading for well reasoned responses.

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